Ryan Dolan Rodriguez
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EDUCATION

University of Notre Dame | Notre Dame, Indiana — Expected Graduation Date: May 2028
Bachelor of Science, Mechanical Engineering| Minor: Engineering Corporate Practice

Engineering Study Abroad | Alcoy, Spain | Universidad Politécnica de Valencia — Summer 2025

ENGINEERING PROJECTS

CubeSat - IRIS Design Team| University of Notre Dame — August 2025 – Present

- Reduced aerodynamic drag uncertainty by 30%+ on flight-critical deployment mechanisms by running CFD simulations to characterize airflow profiles and validate structural load assumptions at 89,000 ft altitude
- Eliminated structural failure risk on a 5.5 lb near-space payload by performing FEA in Ansys to evaluate stress, deformation, and factor of safety under aerodynamic and deployment loads using first-principles analysis
- Reduced per-unit system cost to ~$1,200 by personally fabricating structural components via CNC machining, waterjet cutting, and 3D printing, and coordinating machined aluminum chassis plate production to spec
- Enabled 2.5+ hours of thermal survivability at –10°C by integrating servo controllers, GPS sensors, and attitude sensors into a custom PCB, routing power across three isolated rails for heating, avionics, and actuation circuits
- Validated mission readiness above 89,000 ft by designing and executing a hardware test plan across thermal, structural, and electrical subsystems, presenting results in a formal design review

Satellite Payload Vibration Isolation| University of Notre Dame — August 2025 - December 2025

- Validated a spring-mass-damper model of the SpaceX Falcon Heavy launch environment by matching MATLAB numerical simulations to closed-form analytical solutions across multiple base excitation scenarios
- Designed an optimal vibration isolation system for a 2,982 kg satellite payload by evaluating Moog SoftRide UniFlex isolator configurations across a 5–100 Hz sine environment, minimizing peak acceleration transmitted to the spacecraft

WORK EXPERIENCE

AI Leaders of Tomorrow Internship| Industry Labs, University of Notre Dame — May 2026 – August 2026

- Built a retrieval-augmented AI agent for a $1.3B+ OEM supplier that eliminated manual cross-referencing of engineering specifications and removed the compliance risk of citing an outdated spec revision, achieving 83.5% answer accuracy with zero hallucinations across a 103-question test battery, with every answer traced to a verified document, section, and page.
- Quantified an 80% reduction in spec-lookup time, translating to an estimated 3,000 hours and $200K+ in recovered engineering capacity annually across 60–70 engineers, by modeling the tool's measured performance against current manual-lookup benchmarks.
- Unlocked 42% of a technical document library that had been completely unsearchable, recovering thousands of specifications by processing scanned documents 6x faster.

LEADERSHIP

Society of Hispanic Professional Engineers (SHPE) - Conference Chair | University of Notre Dame

- Spearheaded planning and execution of a 40+ person trip to the national SHPE conference, managing logistics, travel coordination, and a multi-thousand dollar budget across fundraising and chapter financing
- Raised $10,000+ in conference funding by organizing and executing football gameday fundraisers, leveraging Notre Dame's high-traffic game days to maximize revenue and cover full delegation travel and registration costs
- Drove conference preparedness across 40+ delegates by hosting professional development programming, including a chapter-wide resume review and networking workshops

TECHNICAL AND LANGUAGE SKILLS

CAD/CAM: SolidWorks, Autodesk Fusion 360, GD&T

Simulation: Ansys (FEA, CFD), MATLAB

Fabrication: CNC machining, waterjet cutting, 3D printing, Injection Molding

Programming: Python, MATLAB

Languages: English (fluent), Spanish (professional)